

Exercise 03
Convex Optimization

Exercise 1 **Convexity of Sets**

Which of the following sets are convex? Give reasons for your answers.

- 1) $\{x \in \mathbb{R}^n \mid x^\top x \geq 0\}$
- 2) $\{x \in \mathbb{R}^n \mid -x^\top x \geq 0\}$
- 3) $\{x \in \mathbb{R}^n \mid x^\top x = 0\}$
- 4) $\{x \in \mathbb{R}^n \mid \alpha \leq a^\top x \leq \beta\}$ where $a \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$.
Hint: Try to sketch this set for $n = 2$!

Exercise 2 **Quadratic Program**

Consider the optimization problem:

$$\begin{aligned} \min \quad & \frac{1}{2}(x_1^2 + x_2^2 + 0.1x_3^2) + 0.55 x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \\ & x_3 \geq 0 \end{aligned}$$

- 1) Show with the KKT-conditions that $x^* = (0.5, 0.5, 0)$ is a local minimum.
- 2) Is x^* also a global minimum? Explain why, or why not.

Exercise 3 **Non-convex Optimization**

Consider the non-convex optimization problem

$$\begin{aligned} P_1 : \min_{x \in \mathbb{R}} \quad & (x - 2)^2 \\ \text{s.t.} \quad & |x| = a \end{aligned}$$

for $a \in (0, 1)$.

- i) Determine the global optimal solution $x^*(P_1)$
- ii) Relax (reformulate) the optimization problem P_1 such that it becomes a convex optimization problem P_2 for which you can prove that the optimal solution $x^*(P_2)$ is the same as $x^*(P_1)$ for all $a \in (0, 1)$

Exercise 4 **Slater's Condition**

Consider the following convex optimization problem¹

$$\begin{array}{ll}\min_{x \in \mathbb{R}, y \in \mathbb{R}_+} & e^{-x} \\ \text{subj. to} & x^2/y \leq 0\end{array}$$

- i) Does Slater's condition hold?
- ii) Calculate the duality gap.
- iii) Show that the KKT conditions are not respected.

¹This exercise is adapted from *Stephen P. Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.*